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Capstone Planning

Goal

High Level: Create a "light weight" rebalance engine for Vise, primarily to be used to manage fund models.

Motivating Use Case

An RIA firm will provide Vise with a suite of models and would like Vise to "manage" their clients' accounts in accordance with the model they have selected.

To make it more tangible:

- RIA Firm Brandt Wealth Management (BWM) has a suite of models (21 total) of varying risk tolerance. Risk tolerance is denoted with the portion of the model allocated to equity vs fixed income (the 80/20 EQY/FI model is riskier than the 30/70).
- Brandt, an advisor at BWM, has 100 clients.
- Brandt has assigned each of his 100 clients into one of the models
    - Client A's portfolio will be tracking BMW 60/40
    - Client B's portfolio will be tracking BWM 15/85
- Brandt has delivered this information to Vise and told Vise "manage these accounts."
- By "manage these accounts" Brandt wants Vise to trade the securities in each account so that they track the underlying model (which is occasionally

updated by BWM's CIO). Additionally, Brandt expects Vise to be "harvesting losses" in the accounts, when appropriate.

We want to build an engine that can power "manage these accounts".

Components Needed

High level outlining:

1. Build a tax-aware, simulation engine to backtest various methodologies
2. Possibly: define methodology for creating "replacement names"
3. Define triggers for rebalancing portfolio
4. Define rules for carrying out trades
5. Clearly defined recommendation with backup for 2-4
6. Display pretty charts and metrics to aid in decision making
7. Build prototype UI which allows user to test and explore different parameters and methodologies of 2-4

Further Commentary

- The simulation engine is used to inform all other steps so it's important to get this right
- Prototype UI (7) would be valuable and *shouldn't* be much additional effort if step 1 is done well...
- Importance ranking of research recommendations is 4 >3 > 2
- My preference would be
  - Do a complete & rapid, bare-bones version of 1-7
  - Afterward, come back to further explore more sophisticated techniques later.

UI

People respond better to seeing a finished product — especially when they can interact with it — a UI would go a long way to convincing the rest of the Vise team that what you have created here is valuable and working.

A UI for this project might be something like:

- Inputs
    - Send in tickers + weights
    - Asset class tags for securities (possibly multi-level)
    - Taxable Account (True or False)
    - Tax Rates
    - Tax budget (Max allowable capital gains)
    - Date Range
    - Initial Deposit
    - Selection of trading parameters (rebalance triggers, rebalance rules, replacement methodology, etc... just everything you are researching)
- Output
    - Charts
    - Metrics: return, volatility, sharpe, draw down, TE to target model... (everything you can think of + incorporate key after-tax metrics)

Two tools I have used to build UI's with in the past are:

- Streamlit (big fan of Streamit)
- Flask

But keeping everything in a notebook could work if that's where you are most comfortable.

Rebalance Triggers

You all already have a good handle on this, but, for consideration:

- Active share and asset class deviation thresholds
- Calendar based
- Trailing Tracking Error to benchmark or target model
- Projected Tracking Error to benchmark or target model
- % threshold a security or portfolio needs to decline before harvesting
- <Insert whatever else you have found>
- <Insert dynamic approaches proposed in doc>
- <Combinations of the above. Using *and* or *or* conditions. >

Rebalance Rules

- Keep it explainable
- Keep trades within tax budgets
- Do not violate wash sales
- <Ideally, minimize tracking error to target model>

Data

You mentioned you will use WRDS. That is fine for initial tinkering, **but**, we will likely need to switch to Vise-provided data very soon so that everything can be shared with us and easily assimilated into our systems.

See Data - Google Drive. You will find:

- Example Models
- Example proxy lookup.

- ETF & MF Returns (Coming soon...)
- ETF & MF Descriptor Info (Coming soon...)

GitHub

I believe it will be difficult to build clean code that fulfills our desired goals without structuring the project properly in a repository. My preference is that we have a shared repo where we are storing all relevant code. I can conduct a brief GitHub & Git essentials in our meeting if needed.

But, if you all have other methods for coordinating effectively and feel it won't be necessary, let's discuss. I don't want to slow us down by forcing everyone to use Github.

Miscellaneous

- I do not feel any one step should be difficult, but stringing them all together in this time frame could be.
- We have been combining 2 separate goals, which are:
  - For a non-taxable accounts, "explore" various rebalancing methodologies and understand the tradeoffs of each.
  - Demonstrate the efficacy of a pure rules-based approach for accurately tracking target model while still provided tax alpha.
- If you all would like to include an engine powered by optimization as well, by all means, but it's not the priority.
- For research purposes, we can perhaps simplify the scenarios by testing with "simple" portfolios, containing only two holdings. An ETF to represent Equity and one to represent Fixed-Income.
  - Think about the purpose of rebalancing for a simple case. Why is it needed? Is it more about capturing return or managing risk? Which

approach achieves that goal the best? Are there macro-factors impacting the previous question?

- Scoping: it's very important that over the next 1-2 weeks we hone in on what we feel is possible. You need to let me know:
    - What you feel most and least confident on
    - What you feel may take the most or least amount of time
    - Informed from above, if you had to propose cutting something from scope, what would it be
    - Is there anything not emphasized enough right now, you feel should be
- I keep pushing for a UI, but it should not distract substantially from the core goals.
- I like the ideas from your brainstorming doc, and if you would rather dedicate time to exploring these and other ideas as opposed to a UI, let's discuss.